

Entrepreneurial intentions: Applying the theory of planned behaviour

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Understanding and predicting new venture initiation requires research using theory-driven models that adequately reflect the complex perception-based processes underlying intentional, planned behaviours such as new venture initiation. We discuss exactly such a model, widely used in social psychology, and demonstrate its applicability to the entrepreneurship domain. Ajzen's intentions-centred 'theory of planned behaviour' is parsimonious, well grounded in theory, and robustly predicts a wide variety of planned behaviours. Intentions are the single best predictor of such behaviour, both conceptually and empirically. Intentions formation depends on attitudes toward the target behaviour which, in turn, reflect beliefs and perceptions. Intentions-based models of entrepreneurial activity are compatible with existing research results and open new approaches to studying venture initiation. Intentions models allow us to better understand the impact of various antecedents of organizational emergence; identifying not only *what* influences emergence, but also *how*.

Keywords: intentions; organizational emergence; planned behaviour; process models; pre-organizations.

1. Entrepreneurship as intentional, planned behaviour

Planned behaviours such as starting a business are intentional and thus are best predicted by intentions toward the behaviour, *not* by attitudes, beliefs, personality or demographics. In turn, intentions are best predicted by certain specific attitudes. Intentions fully mediate the relationship between attitudes and the target behaviour, even where attitudes may appear to explain behaviour. Intentions entail an enactive cognitive process which serves to channel beliefs, perceptions and other exogenous factors into the intent to act, then to the action itself (Ajzen 1991).

Understanding intentions is particularly useful where phenomena are relatively rare. In dealing with planned behaviours with a very low base rate of occurrence (such as starting a business), intentions offer significant insights into underlying processes (Ajzen 1987). That is, studying intentions gives us valuable insights into new venture initiation, even without observing that initiation. Therefore, we argue that intentions-based models offer a great deal to entrepreneurship researchers.

Entrepreneurial activity clearly represents planned (thus intentional) behaviour. (To illustrate, it is hard to imagine an experimenter ringing a bell followed by a subject then starting a business.) Intentionality is a critical characteristic of emerging organizations, so studying pre-organizational phenomena, including the decision to initiate an entrepreneurial career, is clearly both important and interesting (Katz and Gartner 1988).

Intentions toward a behaviour are absolutely critical to understanding other antecedents (e.g., situational role beliefs), subsequent moderators (e.g., availability of critical resources) and the final consequences of that behaviour (e.g., new venture initiation); intentions also offer significant predictive validity (Sheppard *et al.* 1988). A striking characteristic of intentions-based models is how they handle the antecedents of behaviour. Behaviour is best predicted by intentions, which derive from attitudes. In turn, attitudes derive from exogenous influences. That is, attitudes predict intentions which in turn predict behaviour (Ajzen and Fishbein 1980, Ajzen 1987). Intentions thus serve as a mediator or catalyst for action. Empirically, behaviour is only weakly predicted by attitudes toward that behaviour or by other exogenous factors which may be situational (e.g., employment status or informational cues from the environment) or personal (e.g., demographics or personality traits). Entrepreneurship research typically tests the impact of exogenous factors on entrepreneurial activity and typically finds relatively small effects.

This is a crucial point: exogenous influences usually affect intentions and behaviour indirectly through attitude changes, not directly (Ajzen 1987, Bagozzi and Yi 1989). As Figure 1 suggests, only intentions directly affect behaviour, while attitudes affect intentions. Exogenous influences either (a) drive attitudes or (b) moderate the relationship between intentions and behaviour (i.e., facilitate/inhibit the realization of intentions). Intentions are specific to the person and the context, while exogenous factors are generally either person or situation variables. Thus, it is unsurprising that the influence of exogenous factors is usually indirect. It is also unsurprising that prior research typically finds that exogenous factors only weakly predict entrepreneurial activity. However, the intentions-based approach offers testable, theory-driven models of how exogenous factors affect attitudes, intentions and behaviour.

Accepting the reasonable premise that starting a business is intentional entails accepting some significant implications for research. Stimulus-response models simply do not apply to intentional behaviours. Processes are less well understood retrospectively. Rather, they are best studied prospectively. We thus need testable, theory-driven process models of entrepreneurial cognition which focus on intentions and their perceptual bases.

Social psychology offers us robust, parsimonious models of behavioural intentions which show considerable explanatory and predictive value for a wide range of behavioural phenomena. Martin Fishbein and Icek Ajzen have developed (and continue to develop separately) a theoretical framework that specifically maps out the nature of the process underlying intentional behaviour. The latest version is Ajzen's theory of planned behaviour. This paper argues the merits of this specific model which consistently has exhibited significant predictive validity, typically explaining 30% of future behaviour (Ajzen 1991).

Empirical robustness without conceptual soundness is not progress. Intentions-based models derive their success from two sources. First, planned behaviour is intentional. These models are thus congruent with what we already know about human cognition (and driven by sound theory). Second, they afford a better understanding of hypothesized influences on entrepreneurial behaviours. That is, this model not only assists in understanding what hypothesized influences are significant antecedents to new venture creation. It also provides insight into how they influence organizational emergence.

Intentional processes are highly sensitive to initial conditions (e.g., founders' effects). Intentions establish key initial characteristics for new organizations and reflect founders' critical assumptions about the venture's concept and its social context (Bird 1988, Katz and Gartner 1988, Carsrud 1989, Carsrud and Sapienza 1993). Initial choices by

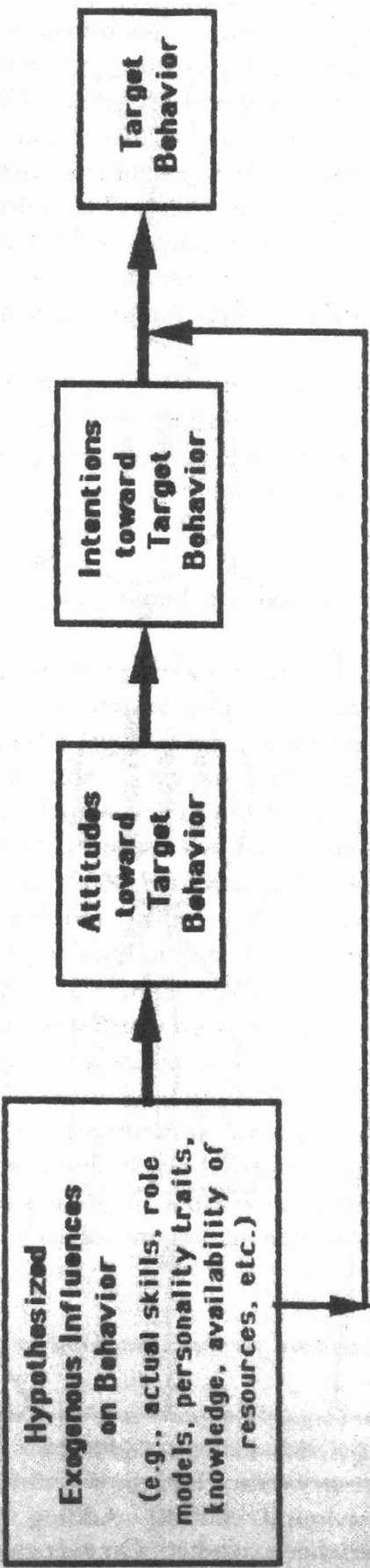


Figure 1. The basic intentions-based process model

business founders have significant repercussions that persist long after firm emergence. Thus, entrepreneurial intentions are crucial to understanding the overall process of organizational emergence (Bird 1988, Boeker 1989, Schoonhoven and Eisenhardt 1990).

All of this argues strongly for testing an intentions perspective to entrepreneurship research. Despite this, entrepreneurship research offers few prospective, process-based studies. The notable exceptions (Long and McMullan 1984, Carsrud 1989, Teach *et al.* 1989, Guth *et al.* 1991) assume intentionality, though none uses an explicit theory-based model of intentions. The empirical robustness and practical utility of existing intentional models derive from the (a) cognitive centrality of intentions in planned behaviours and (b) models' conceptually sound theoretical bases which fully recognize perception-driven process phenomena. Intentions-based models address critics of entrepreneurship research who decry *ad hoc* static models fraught with hindsight biases and censored samples.

Section 2 now examines one such model which offers considerable potential to the study of organizational emergence. Section 3 reviews persuasive evidence supporting such models, Section 4 discusses conceptual and measurement issues for this model, and Section 5 discusses research implications.

2. The theory of planned behaviour

Social psychologists and marketing researchers have long used the Ajzen-Fishbein models of behavioural intentions with great success in practical applications and in basic research. These cognitive models are consistently robust and replicable in predicting behaviour and intentions, including success in studies of career-related behaviours. These models have been widely used in practical applications from health care (weight loss, quitting smoking) to safety (seat belt usage) to marketing (coupon usage) (Ajzen and Fishbein 1980, Ajzen 1987, Sheppard *et al.* 1988). Research in entrepreneurship has largely ignored these models for less robust and less predictive approaches using personality traits, demographics or even attitudinal approaches.

The theory of planned behaviour (see Figure 2) specifies three distinct attitudinal antecedents of intention, each drawn from existing theory and prior evidence. Two of these reflect the perceived desirability of performing the behaviour; personal attitude toward outcomes of the behaviour and perceived social norms. The third, perceived behavioural control, reflects subjects' perceptions that the behaviour is personally controllable; it also reflects the perceived feasibility of performing the behaviour and is related to subjects' perceptions of personal situational competence (i.e., self-efficacy). The model further specifies testable antecedents of each of these attitudes.

2.1. Empirical evidence for the Ajzen-Fishbein framework

A review of empirical tests (e.g., Sheppard *et al.*'s 1988 meta-analysis) indicates that intentions successfully predict behaviour and attitudes successfully predict intentions. Sheppard *et al.* (1988) report an average R^2 around 0.3 for the theory of reasoned action (i.e., no measure of behavioural control). Adding perceived behavioural control consistently increases the variance explained. On average, attitudes explain over 50% of the variance in intentions and, on average, intentions explain about 30% of the variance in behaviour (Ajzen 1991). Explaining 30% of the variance in behaviour compares favourably to the 10% typically explained by trait measures or by attitudes (Ajzen 1987).

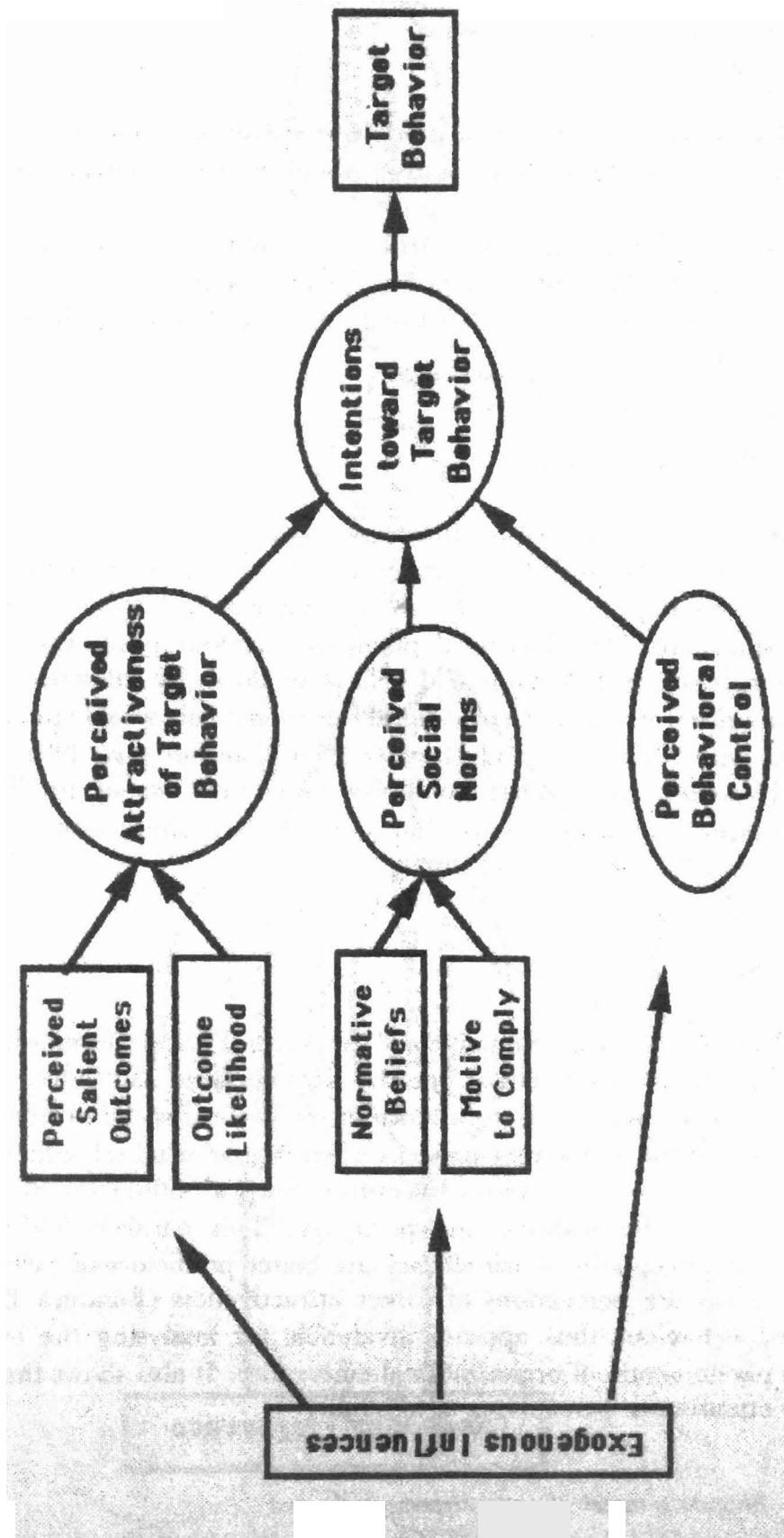


Figure 2. Ajzen's theory of planned behaviour

Intentions fully mediate the influence of attitudes on behaviour. Results are robust to even major misspecifications as long as intentions are properly measured (Bagozzi and Yi 1989).

We discuss conceptual and measurement issues in more detail in Section 4. First, in Section 3 we review empirical evidence supporting the application of intentions-based models to research organizational emergence.

3. Evidence for intentional models of entrepreneurship: (what we know (and don't know) about entrepreneurial intentions)

Few entrepreneurship studies explicitly consider intentions-based models. However, some existing research indicates this would be a fruitful approach. This includes (a) research on entrepreneurial role models, (b) a recent pilot study and (c) Shapero's model of the entrepreneurial event.

3.1. Role model research

Research evidence on entrepreneurial role models supports the application of a model of planned behaviour to organizational emergence. For example, intentions help explain conflicting research findings on the effects of role models and mentors on eliciting entrepreneurial behaviours. The theory of planned behaviour posits that exogenous influences on entrepreneurial intentions and behaviour do so by influencing attitudes (not directly). The existence of an entrepreneurial role model only weakly predicts future entrepreneurial activity (Brockhaus and Horwitz 1986, Carsrud *et al.* 1987, Scott and Twomey 1988). However, role models' subjective impact is a strong predictor. Role models affect entrepreneurial intentions, but only if they affect attitudes such as self-efficacy (Krueger 1993, Scherer *et al.* 1989).

3.2. Results of a pilot study

Preliminary results from one experimental test of the Ajzen-Fishbein framework suggest entrepreneurial intentions do depend on the hypothesized attitudes. Perceived behavioural control explained the most variance in the pretest; social norms explained the least amount of variance. Different patterns of entrepreneurial self-efficacies might explain intentions toward different types of ventures. Such a finding would have great applied predictive utility for training and pedagogy. This parallels findings which strongly suggest that perceptions of self-efficacy are better predictors of career interest and actual choice than are perceptions of career attractiveness (Bandura 1986). The theory of planned behaviour thus appears invaluable for analysing the intentional, perception-based phenomenon of organizational emergence. It also shows that we need to pay particular attention to perceptions of feasibility.

3.3. Shapero's model of the entrepreneurial event

One existing intentionality-based model is Shapero's model of the entrepreneurial event (1982). As in Ajzen-Fishbein models, intentions are a necessary precondition for the target behaviour, new venture initiation (see Figure 3). For Shapero, intentions require

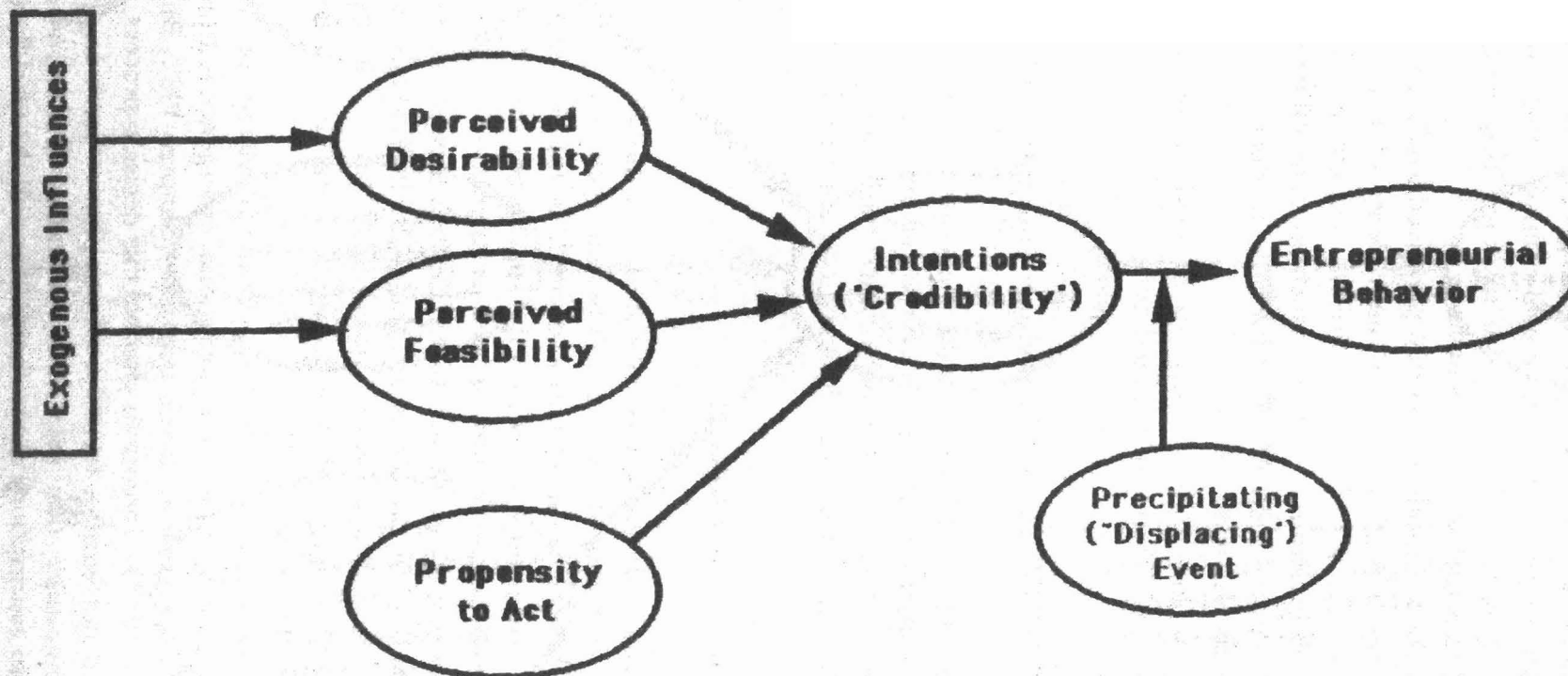


Figure 3. Shapero's model of the 'entrepreneurial event'

that founders perceive entrepreneurship as a 'credible' career alternative. 'Credibility' depends on perceptions that the venture is both desirable and feasible. Similarly, exogenous influences such as unemployment, divorce and other external events operate indirectly through attitudes of perceived desirability and feasibility. Perceived desirability is akin to Ajzen-Fishbein measures of the behaviour's personal attractiveness; perceived feasibility corresponds to perceived behavioural control. (Shapero's 'propensity to act' adds a volition-based attitude. Shapero also includes 'displacement', a precipitating event that triggers the actual intended behaviour.)

A recent test of Shapero's model shows that perceived feasibility, perceived desirability and propensity to act are all significant antecedents of entrepreneurial intentions. Exogenous factors such as prior exposure to entrepreneurial activity influence perceptions of desirability and feasibility, which in turn influence intentions toward entrepreneurial behaviour. That is, attitudes fully mediate the impact of exogenous factors upon intentions (Krueger 1993).

It is appropriate at this point to turn to detailed discussion of conceptual and measurement issues in the theory of planned behaviour.

4. Theory of planned behaviour: concepts and measurement

Figure 4 depicts a testable model based on the theory of planned behaviour and adapted for application to research into entrepreneurial behaviours. As noted above, intentions (not attitudes) best predict planned behaviours; attitudes predict intentions. The following sections discuss critical issues for each construct in the model.

4.1. *Intentions and behaviour*

4.1.1. Conceptual issues: Philosophers who study the question of 'action' agree with psychologists on the centrality of intentions in human agency. Intentions reflect both the desire to act and the belief that one will act and should be the proximate cause of action or trying to act (Brand 1984). Intention is thus the appropriate unit of analysis, particularly where behaviour requires opportunity as well as intention. Intentions are a cognitive structure including both goals (ends) and plans (means), though goals typically crystallize in subjects' minds before the plans to reach the goals. Individuals identify goals before identifying plans to reach those goals (Ajzen 1987, Tubbs and Ekeberg 1991). For instance, evidence shows that people tend to decide to start a business *before* they decide what type of business (Brockhaus 1987).

Entrepreneurial intentions also reflect the founders' vision (or 'unwritten business plan') of the emerging organization. Organizational emergence is a process consisting of a series of purposeful, perception-driven decisions (Shapero 1982, Bird 1988, Katz and Gartner 1988, Gartner 1989). Intentions channel this decision process, whether formal or informal. (One need not write a formal business plan to have an intention to start a business or to demonstrate that intention.) Founders' intentions and attitudes are inextricably bound to assumptions about themselves and the world around them. These in turn depend on perceptions and experiences (Schein 1983, Weick 1979).

Entrepreneurial behaviours entail relatively complex linkages between intention and action with longer time lags. Even very strong intentions need not lead to actual behaviour (Triandis 1967, Katz 1989). This argues for considering not only temporal issues (Bird 1991) but also factors which may precipitate (e.g., 'displacement' and 'propensity to act'), facilitate or inhibit entrepreneurial behaviour.

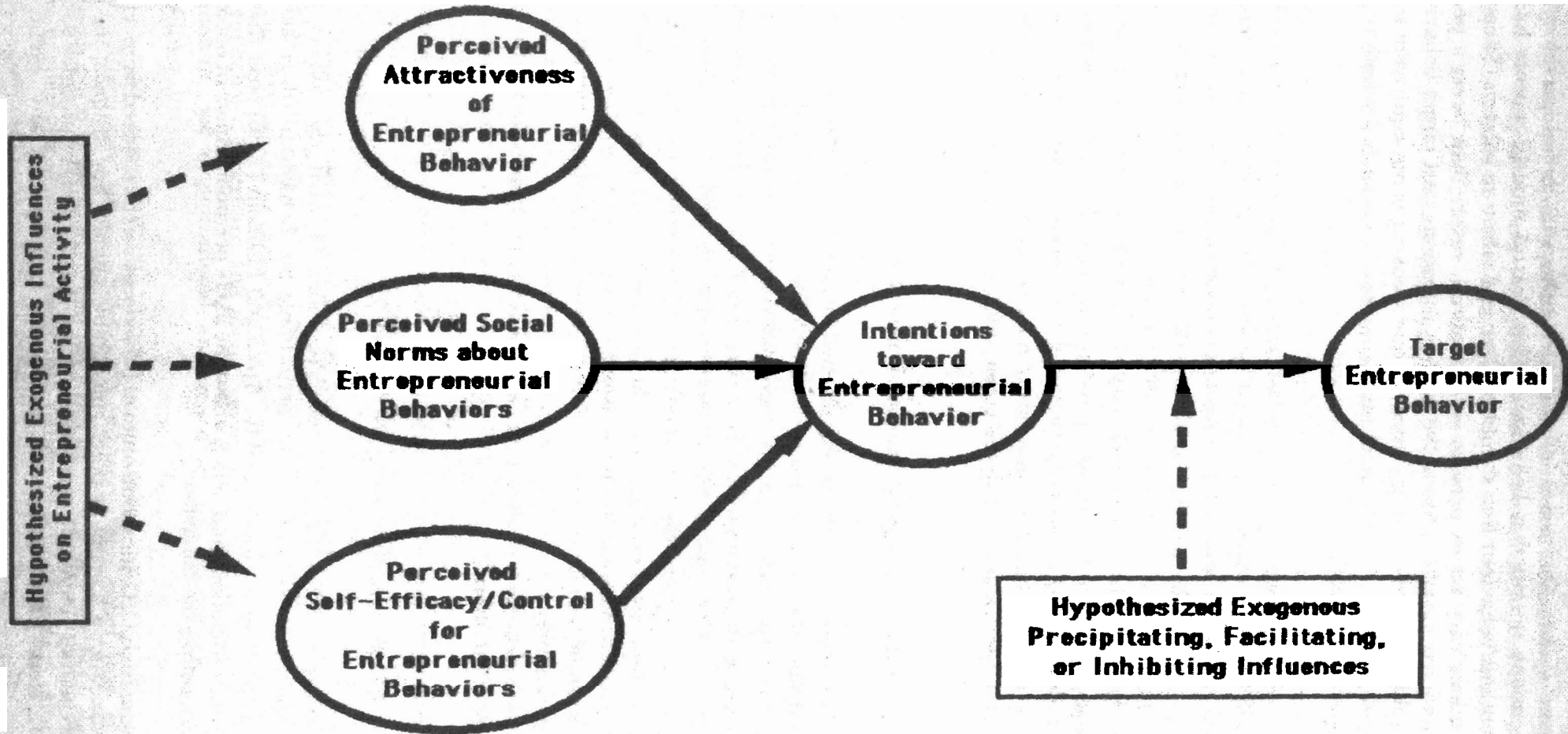


Figure 4. Intentions toward entrepreneurial behaviour: the theory of planned behaviour (simplified)

4.1.2. Measurement issues: Discussion of a target behaviour is inseparable from discussion of intentions toward that behaviour. This class of models appears robust; however, the intentions/behaviour link should be clear and salient to subjects (Sheppard *et al.* 1988). Intentions vary across persons and situations, rather than being a person variable or a situational variable (Ajzen 1987), thus intentions and target behaviours should be comparable in specificity (e.g., intent to quit smoking is not equivalent to the intent to try to quit smoking). Finally, multiple measures of constructs are both typical and useful (Sheppard *et al.* 1988).

4.1.3. Sampling: As suggested by meta-analysis (Sheppard *et al.* 1988) and by Katz and Gartner's (1988) call to examine pre-organizations, the sensitivity of intentional processes to initial conditions argues for studying phenomena *before* they occur and for inclusion of non-intending subjects. We too often ignore those who do not intend to start a business, despite the oft-cited interest in differentiating entrepreneurs from non-entrepreneurs. (Other parallels: clinical studies try to start with subjects prior to the onset of illness; geologists try to study volcanoes prior to eruption and include volcanoes in various stages of dormancy.) Sensitivity to initial conditions means the process is inherently non-linear. Sampling only entrepreneurs thus introduces a bias that will certainly censor data and do so with unpredictable results.

4.2. *Attitudes: antecedents of intention*

4.2.1. Conceptual refinements: The theory of planned behaviour identifies three attitude measures as critical antecedents. Two measure subjective expectations about the personal attractiveness of the target behaviour: (a) personal attitude toward performing the behaviour (presumed to depend on the likely impact of salient outcomes from the target behaviour) and (b) perceived social norms (presumed to depend on perceived social pressures which favour or oppose the behaviour). The third attitude measure, perceived behavioural control, represents optimism that the target behaviour is actually doable or achievable. Perceived behavioural control is closely akin to Bandura's notion of self-efficacy, a consistent, significant predictor of career-related attitudes and intentions (Bandura 1986, Ajzen 1991).

Let us now turn to discussion of conceptual and measurement issues for the three key attitude constructs of the model.

4.3. *Attitude toward performing the behaviour*

4.3.1. Conceptual issues: This concept taps perceptions of what the subject finds personally desirable or undesirable about performing the behaviour (thus akin to expectancy). As a check on construct validity, Ajzen and Fishbein (1980) posit that this attitude depends on expectations and beliefs about the likely personal impact of outcomes deriving from performing the behaviour.

4.3.2. Measurement issues: Critical outcomes of behaviours can be solicited from a focus group drawn from either a holdout or other comparable sample. For instance, the pilot study on entrepreneurial intentions mentioned earlier found critical outcomes such as personal wealth, stress, autonomy and community benefits. In any event, the

entrepreneurship literature offers an ample number of potential outcomes whose impact on entrepreneurial attitudes and intentions is testable through this model.

4.4. *Perceived social norms*

4.4.1. *Conceptual issues:* This construct taps subjects' perceptions of what the important people in their lives think about performing the behaviour (e.g., peer pressure, family expectations, etc.); thus it is driven by normative beliefs. As a check on construct validity, these subjective social norms should depend on the expected support or non-support of significant others. Interestingly, social norms are less relevant to intentions for subjects with a highly internal locus of control (Ajzen 1987) or who exhibit a strong orientation toward taking action (Bagozzi *et al.* 1992).

4.4.2. *Measurement issues:* The critical choice is identifying the important social influences (e.g., parents, 'significant other', friends, etc.), including any 'role model' or 'mentor'. Research into the personal networks of entrepreneurs has often focused on flows of resources (e.g., information); less research has looked at social norms and values provided by network members (Shapero 1982, Johannisson 1990).

4.5. *Perceived behavioural control and self-efficacy*

4.5.1. *Conceptual issues:* Perceptions of desirability and feasibility underlay career-related decisions, including entrepreneurial ones. Feasibility perceptions have been consistently more important in predicting behaviour, particularly for behaviours where control is problematic (e.g., Madden *et al.* 1992). People tend to choose among desirable career options based on their perceived feasibility (Driver 1988).

Perceived behavioural control overlaps considerably with Bandura's notion of perceived self-efficacy (Ajzen 1987). Self-efficacy is an extremely powerful predictor of goal-oriented behaviour in a wide range of domains. Correlations between self-efficacy and career intent consistently range from 0.3 to 0.6 (Bandura 1986), results that compare quite favourably with most predictors currently used in entrepreneurial research such as locus of control (Carsrud and Pate 1990). Self-efficacy influences both outcome expectancies and aspiration levels for target behaviours (Locke *et al.* 1984). These in turn increase the strength of one's intentions (Bandura 1986).

Despite its proven robustness, self-efficacy theory has been largely ignored by management and entrepreneurship researchers despite its ability to predict both general and specific behaviours (Gist 1987). For instance, we have already noted that the impact of role models on entrepreneurial intentions depends on whether they affect self-efficacy. In addition, self-efficacy has been associated with opportunity recognition (Krueger and Dickson 1993) as well as career choice. Self-efficacy should be a particularly useful tool in the researcher's repertoire and entrepreneurship researchers seeking a 'psychological explanation' for organizational emergence should examine the role of perceived self-efficacy.

4.5.2. *Measurement issues:* Self-efficacy measures the belief that one can actually execute a given behaviour ('I can get 8 out of 10 calculus problems right') Gist (1987) recommends a global measure based on a battery of self-efficacies at critical competencies

as determined by experts or focus group. Researchers could use the individual self-efficacies to test within-subjects differences in preferred type of venture.

4.6. *Exogenous factors*

4.6.1. *Conceptual and measurement issues:* Generally, exogenous factors either influence attitudes or the intentions-behaviour relationship. Rarely do they directly affect intentions or behaviour. Exogenous factors are typically either person variables (traits, demographics) or situation variables (economic climate); their effects on person-situation variables such as intentions are likely to be indirect. If exogenous factors influence intentions and behaviour by changing attitudes, we need only specify which externality plausibly influences which attitude. For instance, we might test whether the perceived availability of resources such as financing affects perceptions of self-efficacy. Similarly, we can examine exogenous factors that plausibly facilitate or inhibit how intentions are realized. For instance, we can test Shapero's 'precipitating event' or Triandis's 'resource availability' as a moderator of intentions-behaviour links.

In the long run, exploring the mechanisms underlying these relationships (e.g., via image theory or scripts) should prove both interesting and enlightening (Bird 1988, Cantor 1990). For instance, how exactly do prescriptions of resource availability arise and how do they translate into changes in attitudes such as self-efficacy? Proven antecedents of self-efficacy (and thus intentions and behaviour) include actual 'hands-on' mastery (e.g., acquiring business skills) and vicarious learning (e.g., prior exposure to entrepreneurial activity).

Similarly, as we are dealing with a process phenomenon, we might focus on the temporal dimensions of intention formation. What is the 'pacing' or the 'rhythm' of how a business idea becomes reality (Bird 1991)?

5. **Implications**

The theory of planned behaviour's demonstrated robustness offers much to research on the emergence of new organizations. Theory-based intentional models successfully reflect the nature of the intentional, perception-driven process phenomenon of organizational emergence and suggest three important implications for researchers.

- 1 Intentions are the best predictor of planned behaviour; attitudes influence behaviour through effects on intentions. Intentions and attitudes depend on the situation as well as the person and predict behaviour better than strictly individual (personality) or situational (employment status) variables. Also, given that new venture initiation is a relatively infrequent occurrence, studying intentions offers valuable insights into the process, even where we cannot observe initiation.
- 2 The versatility and robustness of intentions-based models should hearten proponents of process models of entrepreneurship and should encourage further process-based research. Much as understanding the nature of intentional behaviour helps explain why many entrepreneurs decide to start a business prior to deciding what type of business to start, increased use of process models should help us lay to rest additional 'entrepreneurial folklore' (Brockhaus 1987).
- 3 Perhaps the most critical distinction is that situational and individual difference variables are exogenous factors that influence entrepreneurship indirectly, not

directly. They do so only in so far as they affect attitudes or the intentions-behaviour relationship. For instance, role models affect entrepreneurial intentions only if they affect attitudes such as self-efficacy, explaining why Carsrud *et al.* (1987) found less impact by role models on entrepreneurial intentions than Scherer *et al.* (1989). Intentions-based frameworks offer a mechanism to assess the relative impact of various hypothesized exogenous influences (e.g., perceptions of resource availability) on intentions and, ultimately, behaviour.

5.1. *Implications for practice and teaching*

These models have already yielded a wide variety of useful practical applications where behaviour is intentional (see review articles such as Sheppard *et al.* 1988). Entrepreneurship teachers, consultants and entrepreneurs should equally benefit from a better understanding of how intentions are formed and how their beliefs, perceptions and motives coalesce into the intention to start a business, especially a particular type of business. Teaching people about the realities of entrepreneurship may increase their entrepreneurial self-efficacy, but simultaneously decrease the perceived desirability of starting a business. As educators, we might use this model to better understand our students' motivations and intentions, and thus to provide better training. Policy makers also benefit from understanding that a policy initiative will affect business formations only if that policy is perceived in a way that influences attitudes (or intention-behaviour relationships).

5.2. *Implications for entrepreneurial research*

Intentions models such as the theory of planned behaviour demonstrate great utility to social psychologists and thus offer considerable potential for entrepreneurship researchers. For instance, researchers might use this model to analyse how the process of doing a business plan or entrepreneurial training affects intentions.

5.3. *Methodological implications*

Intentions-based models address several common criticisms levelled at entrepreneurship research. Instead of the more typical *ad hoc* models of new venture initiation, critics suggest a need for careful testing of theory-driven models (Carsrud *et al.* 1987, Low and MacMillan 1988, Gartner 1989). Another significant criticism is the field's preference for retrospective and *post hoc* research methods. Organizational emergence is an intentional process; let us use models congruent with that reality. We should evaluate the entire emergence process including not only successful and unsuccessful entrepreneurs, but also those who change their minds or whose intentions are unrealistic. The reward for including these is a better understanding of the mechanisms by which exogenous factors influence emergence. Understanding intention helps us better understand where ideas for a business venture come from and how the venture becomes a reality (Carsrud 1989).

For intentional behaviour, hindsight is not perfect. If organizational emergence is simply a response to a given set of stimuli, then biased recall of prior attitudes and experience is less critical. However, if venture emergence is indeed an interactive process, then single 'snapshots' of that process can be misleading. Researchers may

accept spurious correlations (and spurious non-correlations) fuelled by biased recall measures and inappropriately censored samples. *Ad hoc* models unpredictably amplify these biases by grounding models on equally spurious demographic relationships (Carsrud *et al.* 1993) and thus lead to untenable conclusions about entrepreneurial development.

5.4. *Implications for management research*

This model also demonstrates how studying entrepreneurs could be more broadly useful to management research. Entrepreneurs are highly visible exemplars of organizational enactment and cognitions of entrepreneurs should be relatively easier to track than the collective thought of organizations (Guth *et al.* 1991). What we have already learned from studying entrepreneurs' intentions already adds to a broader understanding of intentions. Shapero's notions of a 'displacing' (precipitating) event and of a 'propensity to act' already add to our understanding of the intention-behaviour relationship in general. This addition of volition (and thus action) could be a unique contribution of entrepreneurship research to broader issues of strategic decision making. If social norms affect a behaviour's perceived desirability, measures of social capital (e.g., Johannisson 1990) may significantly affect that behaviour's perceived feasibility.

5.5. *Future research directions*

Research has studied many aspects of new venture plans. Intentionality argues instead for studies on the process of planning: how do plans evolve from an idea to a well-developed business concept (Carsrud 1989)? We might try a schema-based approach (Bird 1989, 1991); image theory is also intentionality-based (e.g., Cantor 1990). Martin's (1985) model offers an interesting menu of exogenous factors that represent potential testable antecedents (from traits, demographics and skills to social, cultural and financial support).

Research into entrepreneurial networks suggest that we might find that the social norms of network members are more significant than those of family and friends. We could model these intentions of venture capitalists toward investing in a given company. (Do their most influential perceived social norms come from within the venture capital community? What self-efficacies influence perceptions of a potential investment's feasibility?) The relative importance of different attitudes on intentions will likely differ across cultures; intentions-based models let us explicitly test for such differences (Triandis 1967). Finally, we can extend use of intentions models to post-emergence strategic decisions such as the decision to exit.

Within-subjects measures are especially appropriate methodologically in intentions research (Tubbs and Ekeberg 1991). One may intend to 'start a business', but what *kind* of business? We could test whether specific self-efficacies predict intentions toward specific types of ventures (e.g., measure attitudes and intentions toward more specific new ventures such as high growth versus no growth, craftsperson versus opportunist, prospector versus defender, high tech versus low tech).

We could test alternative versions of this framework such as Bagozzi's 'theory of goal pursuit' and 'theory of trying' (Bagozzi and Warshaw 1990). Prior behaviour or experience may explain additional variance in intentions and behaviour (Ajzen 1991). Finally, there is growing interest in entrepreneurial careers and intentions-based models

offer testable hypotheses such as whether Schein's 'career anchors' significantly influence perceptions of self-employment desirability (Katz 1992).

The theory of planned behaviour is parsimonious, theory-driven, robust in both explanation and prediction, and has practical utility. Work is already under way to apply these to organizational emergence (and other pre-organizational phenomena) on both a between-subjects and a within-subjects basis with encouraging preliminary results. Intentions models are compatible with the findings of much existing entrepreneurship research, allowing researchers and practitioners to understand the impact of hypothesized influences on organizational emergence: not only what influences emergence, but also how it does so. Finally, such cognitive models may offer the opportunity to use entrepreneurs to help explain broader intentional phenomena of concern to all behavioural scientists.

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